

Phase 4 Report

i.) Overall Description

“Vault Escape” is a 2D arcade-style game. The player plays as the thief manoeuvring through the vault. The game is a combination of action, puzzle-solving and stealth experiences to reach one main goal - escape. Players must go through several maze-like levels and collect valuable rewards on the path. Higher levels introduce new obstacles such as guards, cameras, dogs and lasers. There is a time limit for each level, so players need to be quick to collect all rewards and escape the vault! This combination of fast movement and strategic thinking allows the player to immerse as a real robber.

ii.) Initial Design

The initial design of our game “*Vault Escape*”, was designed as a 2D arcade-style game that introduces players to a mix of action, stealth, and puzzle-solving mechanics. The implementation of “*Vault Escape*” is categorized into three major areas as follows:

1. Stealth Mechanics

Players are challenged to avoid detection by patrolling guards, surveillance cameras and other security systems before time runs out. To add tension, the game imposes a time penalty if the player comes into contact with the security entities.

2. Puzzles and Problem Solving

The entrance door, which also behaves as the exit, only opens upon collecting all the valuables that spawns randomly throughout the map.

iii.) Design of Final Product

The final design inherited a lot from its origins, but some changes were made. Those are:

1. Additional enemy types

Since creating just guards and cameras made us feel that the level difficulty changes would not affect the player properly, we introduced new enemy types : lasers and dogs. It made level progression more noticeable.

2. UI design changes

We stuck with a slightly different style of the game. This made it more arcade and visually appealing.

3. Scoring system and best scores

We introduced the new scoring system, where basic rewards give both value and progress towards an exit opening, while bonus rewards give you plain score bonuses. This made the scoring system more user friendly and collecting rewards became more attractive. Moreover we created a system that allows the player to store their best results in the best scores table in the menu, which made the game even more challenging and engaging.

iv.) Lessons Learned

1. Modular and flexible design

Being able to modify our initial plan and adapt to the changes made, while maintaining clean and modular code allowed us to progress much faster. This also reduced the amount of problems that we could have potentially faced.

2. Team collaboration

Communication is the key. Conducting meetings and talking about existing problems allowed us to distribute the tasks properly. This enhanced efficiency and quality of our work.

3. Efficient debugging and testing

Debugging the whole way throughout the project allowed us to minimize the amount of failures we faced on the testing phase. Creating the proper test set that eliminated any room for error in the project was extremely important to make our system robust.

4. Continuous commits

Using the version control system (Git) allowed us to simplify the process of working together. Ability to look through the commits history allowed us to identify the issues made before and fix them in a quick manner.

Video Link: <https://youtu.be/UI2wqSLGMgY>

Other Gameplay Highlights / Features:

Diagonal Movements:

Unlike a typical 2d indie game, *Vault Escape* offers dynamic movement and allows players to move diagonally throughout the map. The game has also implemented WASD keys along with the arrow keys to avoid control confusion.

FPS (Frames Per Second):

Instead of "tics" *Vault Escape* runs in a FPS based clock system that allows for smooth flowing gameplay.

Cameras:

Cameras increase the speed of the enemy entities if touched by the player.

General Enemy Entity Interaction with Player:

If the player touches any of the enemies, the timer decreases, leaving less time for the player to defeat the level.